

CommuteBuddy



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About CommuteBuddy

CommuteBuddy is a smart web app designed to plan commuters' daily journeys more efficiently.

It integrates real-time public transport data, carpark information, and crowd insights into one seamless platform, allowing users to make informed travel decisions.



Product Features – Smart Commute / CommuteBuddy Web App:

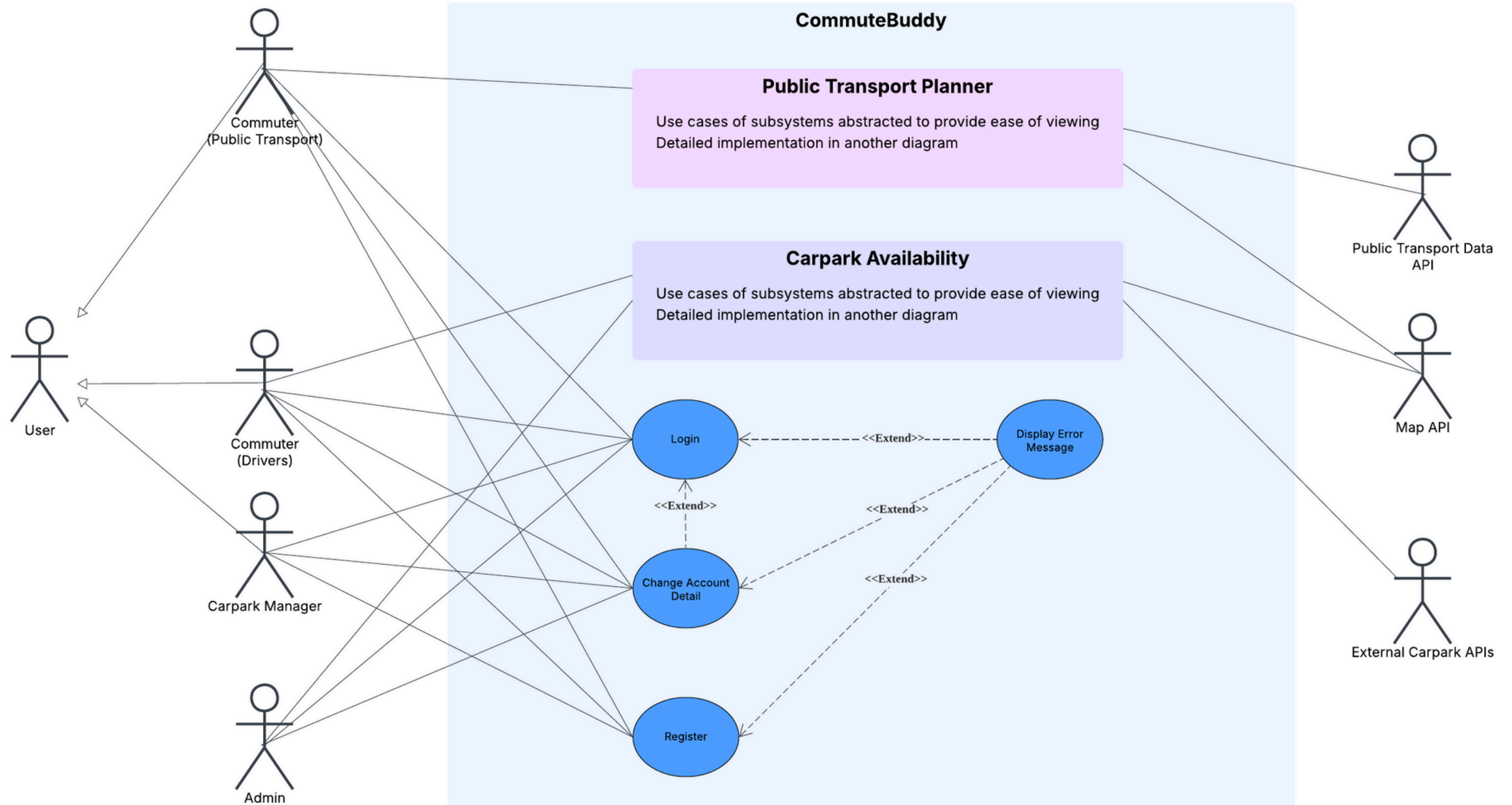
- **Public Transport Planner** – Provides real-time bus arrival times and MRT crowd density to suggest the most comfortable and efficient routes.
- **Carpark Searcher** – Displays real-time carpark availability and locations, with filters to show the total number of lots and the number of available lots within the user-chosen radius. It also allows managers to add non-HDB carparks through a custom API using simulated data for testing.

What sets us apart :

CommuteBuddy stands out by integrating both public and private transport into one platform - offering:

- 1 Focused delivery of public transport information - Simple
- 2 Mrt crowd density data
- 3 Flexible interface for new carpark to be added

Use Case Model



External APIs

Frontend

1. Google Maps

Backend

1. LTA Datamall

- a. Bus Information (Schedule, Services, Stops)
- b. MRT Crowd Density

2. Data.gov.sg

- a. Carpark Availability
- b. HDB Carpark Info

CommuteBuddy – smarter, real-time travel decisions

Live
Demo

Tech Stack

Frontend:

Framework- React.js

Styling- Tailwind CSS

Icons- React icons, SVG

Build Tool- Vite

Backend:

Language - JavaScript, Python, CSS, HTML

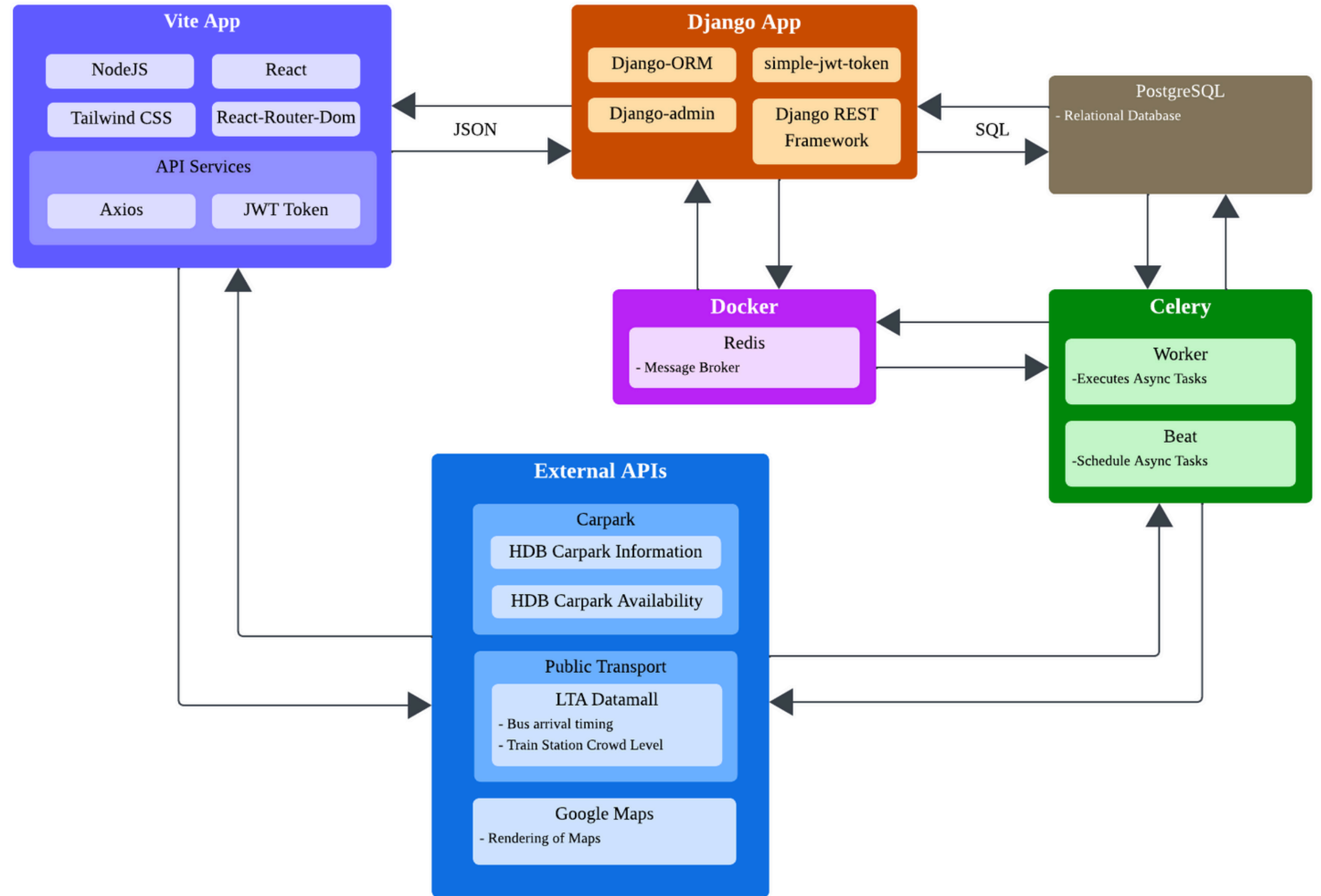
Database- SQLite 3

API integration- Django (fetching), Get and Post

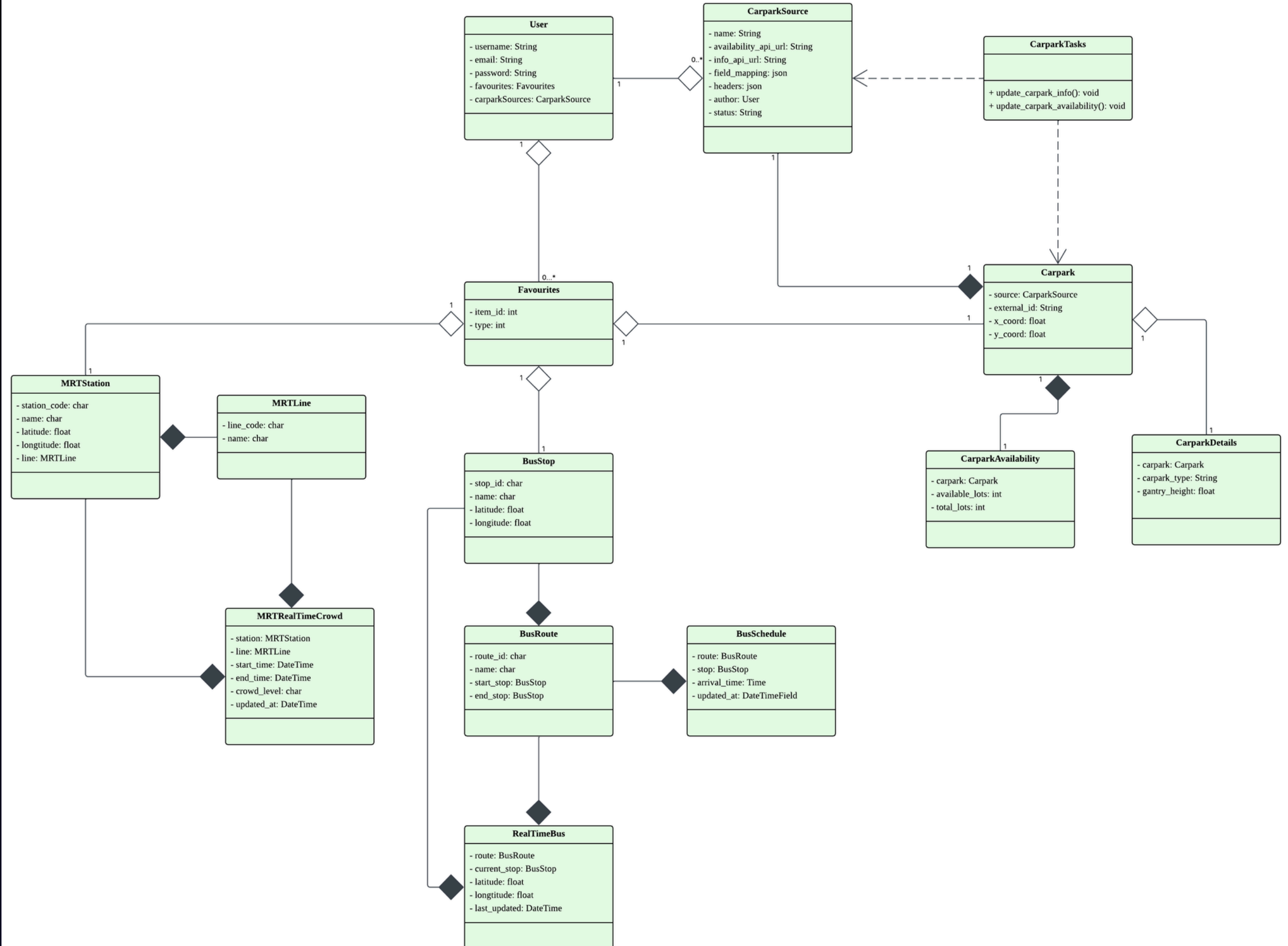
Server tools- JSON for data exchange

System Design

Architecture diagram :



Class diagram :



Software Engineering Practices & Design Patterns



Good SWE Practices

1. Documentation
2. Good Code Practices
3. Reusability & Refactoring
4. SCRUM

Documentation :

Code Consistency:

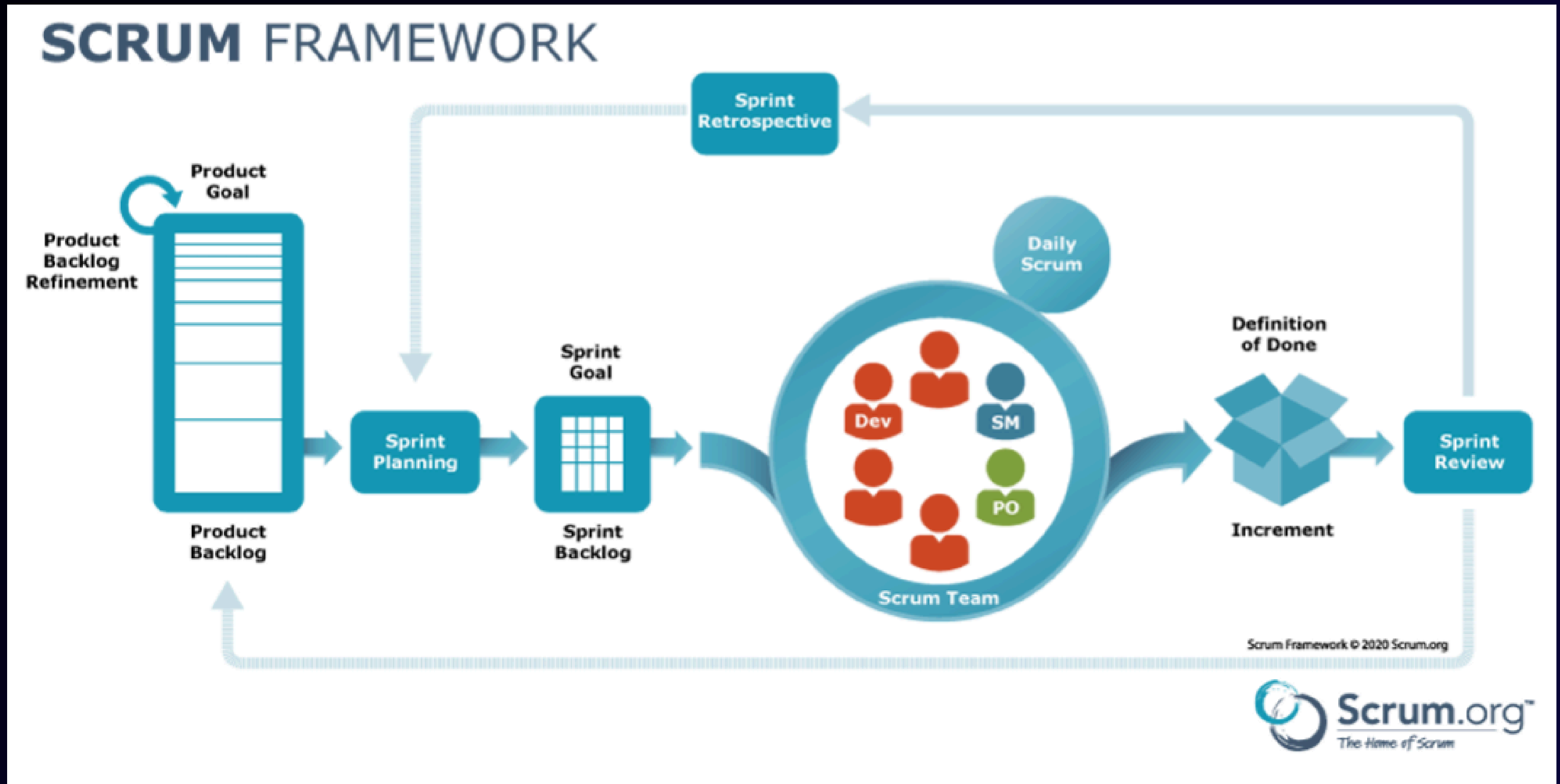
Maintaining a consistent coding style improves readability, simplifies debugging, and ensures easier long-term maintenance.

API Documentation:

Provides clear guidelines for frontend-backend collaboration and supports future scalability and feature integration.

Code Comments:

Helps future developers quickly understand the logic and purpose of the code, enabling smoother updates and enhancements.



Design Patterns

Separation of Concern

- Frontend(Vite) and Backend(Django) Separation
- Allows both teams (frontend and backend) to work independently.
- Flexibility in using modern frontend technologies (React etc.) with Django without tightly coupling them.

Model View Template (MVT)

- Django
- Model – Handles object logic
- View – Handles controller logic
- Template – Not used, replaced with a vite frontend

Observer Pattern

- The frontend uses observer pattern extensively to react to different changes
- Example:
 - Username displayed when login
 - Adding carpark source automatically triggers fetching of data

Challenges:

Lack of Available Public API:

- Lack of API:
 - Train arrival timing
 - Non-HDB carparks
- No single source of carparks

Inconsistent API Format:

- JSON format with different schema and field names
- Challenge to design a one-size fit all API parser
 - Possibly change to allow external API to publish instead of polling from them

Future Plans:

Multi-Language support:

Add translations (e.g. Mandarin, Malay, Tamil) to make the app more inclusive for Singapore's diverse commuter base.

Gamification and rewards:

Introduce badges or points for using sustainable modes of transport (walking, cycling, public transport), encouraging eco-friendly commuting habits.

AI-powered predictive features:

Predict future congestion, carpark availability, or transport delays using AI models trained on historical data

Data Analytics :

Shows data of infrastructure utilization rate to assist urban developers in planning and allocating of resources

Thank You



CommuteBuddy